

Oral 6

Bully Algorithm Viva Questions and Answers

1. What is Bully Algorithm?

Bully Algorithm is a leader election algorithm used in distributed systems.

2. What is the aim of this practical?

To implement Bully and Ring algorithm for leader election.

3. Why is leader election needed?

Leader election is needed to select a coordinator process in distributed systems.

4. What is a coordinator process?

A coordinator is a process that manages and coordinates other processes.

5. What happens when the coordinator crashes?

A new coordinator is elected using an election algorithm.

6. Why is it called Bully Algorithm?

Because the process with the highest priority “bullies” lower priority processes and becomes coordinator.

7. What assumption does Bully Algorithm make?

Each process has a unique priority number or process ID.

8. Which process becomes coordinator?

The active process with the highest priority becomes coordinator.

9. What is an election message?

It is a message sent to active processes to start leader election.

10. What happens if a process receives no reply?

It elects itself as the coordinator.

11. What is timeout T in Bully Algorithm?

Timeout T is the waiting time for receiving a response.

12. What happens if a higher priority process responds?

The current process waits for the higher priority process to become coordinator.

13. What is process priority?

Priority is a unique value assigned to each process to determine leadership.

14. What is a distributed system?

A distributed system is a collection of independent computers working together.

15. What is fault tolerance?

Fault tolerance means the system continues working even if some processes fail.

16. What is the purpose of election algorithms?

To choose a new coordinator after failure.

17. What is process recovery?

Recovery means restarting a crashed process.

18. What happens when a high-priority process recovers?

It may become the new coordinator.

19. What is Ring Algorithm?

Ring Algorithm is a leader election algorithm where processes are arranged in a logical ring.

20. How does Ring Algorithm work?

Election messages circulate around the ring until the highest process ID is selected.

21. Difference between Bully and Ring Algorithm?**Bully Algorithm**

Uses highest priority directly

Faster election

Processes communicate directly

Ring Algorithm

Uses ring structure

More message passing

Processes communicate sequentially

22. What is the purpose of BufferedReader?

It reads user input from keyboard.

23. What is the purpose of arrays in this program?

Arrays store process status (alive/dead).

24. What does System.exit(0) do?

It terminates the program.

25. What is process crash?

A process crash means the process stops functioning.

26. What is process recovery in this program?

A crashed process becomes active again.

27. What is the role of initiator process?

The initiator starts the election process.

28. What is the outcome of this practical?

The current process elects itself as coordinator if no higher process responds.

29. What is the final result of this practical?

Successfully executed all the processes.

30. What are advantages of Bully Algorithm?

- Simple to implement
- Fast leader election
- Automatically detects coordinator failure